

Reinforcement Learning for Epidemic Control in a Large-Scale Agent-Based COVID Simulator

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Abstract—This paper investigates the use of deep reinforcement learning for epidemic intervention planning in a large-scale agent-based COVID-19 simulator modeling individual behavior, social networks, economic decisions, and vaccine development. We train deep Q-learning agents to select public health interventions under delayed and stochastic rewards. The final learned policy reduces average infections by approximately 43% compared to a random-action baseline, while exhibiting multi-stage strategies coherent with real-world public health responses. The results highlight both the potential of RL for adaptive epidemic control and the importance of training stability and early stopping in complex environments.

I. INTRODUCTION

The COVID-19 pandemic showed how difficult it is to organize and decide on public-health intervention decisions amid a crisis. Many existing epidemic models rely on simplified SIR equations and cannot represent the economic or social factors that influence the spread of a disease. In this project, we use a large-scale COVID simulation along with an agent to study epidemic dynamics in a more realistic setting.

The simulator models 10,000 individuals connected through a social network, where each person has unique traits such as risk tolerance, vaccination behavior, spreading potential, and awareness. The model also includes community structure, economic decisions, and a full vaccine research and distribution system.

The main goal of this work is to develop a reinforcement learning agent that interacts with this simulator and learns intervention strategies. Actions include awareness campaigns, mobility restrictions, vaccination allocation, and financial decisions. The learning objective for the agent is to learn complex and intuitive strategies to reduce new infections while managing limited resources and maintaining realistic constraints.

II. BACKGROUND AND RELATED WORK

Reinforcement learning has been increasingly explored as a framework for designing adaptive epidemic control policies, particularly during the COVID-19 pandemic. Most prior work formulates epidemic response as a sequential decision-making problem over simplified compartmental models, where RL agents optimize non-pharmaceutical interventions (NPIs) such as lockdowns or school closures.

Libin et al. applied deep RL to large-scale epidemic control using an age-structured SEIR model of Great Britain spanning 379 districts, employing Proximal Policy Optimization (PPO) to learn school closure policies under budget constraints [1]. Their results showed that RL policies can approximate optimal intervention schedules and outperform static heuristics. Khadilkar et al. extended this framework by incorporating economic costs into a spatial SEIRD model with inter-district mobility across 640 regions in India, demonstrating that adaptive RL-based lockdown strategies outperform threshold-based approaches in heterogeneous settings [2].

Other work has emphasized healthcare capacity constraints. Arango and Pelov proposed a deep Q-learning approach for learning cyclic lockdown strategies that explicitly penalize ICU overload, highlighting RL’s ability to manage trade-offs between public health outcomes and social disruption [3]. Beyond compartmental models, Yang et al. investigated epidemic mitigation on complex social networks using deep RL, modeling individuals as nodes in a contact graph and showing that graph-aware policies significantly reduce infections under resource constraints [4]. To facilitate broader experimentation, Colas et al. introduced *EpidemiOptim*, a Gym-compatible toolbox integrating SEIR and SEIRD simulators with RL algorithms such as DQN and evolutionary methods [5].

While these studies demonstrate the promise of RL for epidemic control, they largely rely on low-dimensional compartmental dynamics and assume a single action–reward cycle per timestep. In contrast, our work integrates RL into a large-scale agent-based COVID simulator that explicitly models individual behavior, social network structure, economic decisions, and vaccine research and distribution. The environment introduces delayed and noisy rewards, multiple actions per decision period, and partial observability, creating a substantially more realistic and challenging control problem that better reflects real-world public health decision making.

III. SIMULATOR ARCHITECTURE

A. Environment

The community is modeled as a social contact network in which each citizen is represented as a node and each social connection as a potential transmission path. The network is generated using a Barabási–Albert preferential attachment

algorithm, a model widely used to approximate real-world social contact networks due to its natural patterns.

Infection is driven by daily contact between infected individuals and their network neighbors. Rather than rely on a uniform probability, the model assigns each daily contact a unique transmission probability derived from the biological, behavioral, and situational characteristics of both parties. Specifically, the probability of infection for a susceptible individual is calculated as the product of:

1. baseline biological probability ($b_p = 0.4$)
2. risk tolerance ($r = 0.7-1.0$)
3. adherence to local restrictions ($q = 0.9$)
4. symptomatic severity ($s = 0.0-1.0$)
5. superspreading potential ($k = 0.3-1.0$)

Each factor is instantiated separately for the susceptible and infected individuals involved in a contact. The resulting probability of an infection event is:

$$P = b_p \cdot r_{\text{citizen}} \cdot r_{\text{infected}} \cdot q_{\text{citizen}} \cdot q_{\text{infected}} \cdot s_{\text{infected}} \cdot k_{\text{infected}}.$$

Following infection, individuals remain contagious for five days before transitioning to a removed state, consistent with observed infectious periods among COVID-19 and related viruses.

B. Gameplay

The simulation places an agent (human or AI) in the role of a local public health authority, tasked with mitigating disease spread under uncertainty. The agent interacts with the system through a set of eleven policy actions, reflecting realistic public health options, broadly categorized into: financial investment, behavioral restrictions, and vaccination strategies. Each action carries a defined cost and introduces nonlinear trade-offs, enabling nontrivial gameplay strategies.

The outcomes of most actions are non-deterministic to further encourage gameplay sophistication and to reflect the practical realities of pandemic response. For example, strict community lockdowns reduce, but do not eliminate, transmission. The availability of vaccines is governed by a semi-random approval timeline. Vaccination acceptance varies at the individual level and is unknown to the agent. These elements form a decision space characterized by incomplete information, delayed gratification, and constantly changing risk.

IV. REINFORCEMENT LEARNING FRAMEWORK

We frame epidemic response planning as a sequential decision making problem in which an agent learns to control the spread of infection through repeated interaction with a complex simulation environment. The underlying agent-based COVID model captures social-network structure, heterogeneous behavioral traits, vaccine research and distribution dynamics, and financial constraints. Reinforcement learning provides a principled approach for discovering intervention strategies that adapt to these evolving conditions.

Formally, the problem is modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where $\mathcal{S}$

denotes the state space, $\mathcal{A}$ the action space, P the stochastic state transition dynamics, r the reward function, and γ the discount factor. At each decision step t , the environment produces a state $s_t \in \mathcal{S}$ summarizing epidemiological, economic, and logistical indicators. The agent selects an action $a_t \in \mathcal{A}$ corresponding to a specific intervention available within the simulator. The environment transitions to a new state s_{t+1} according to the stochastic dynamics of infection transmission, individual behavior, and vaccine progress. The agent then receives a reward r_t that reflects the quality of the chosen intervention.

Episodes continue until one of three termination criteria is reached: (1) active infections reach zero, (2) the susceptible population is fully depleted, or (3) a maximum simulation length is exceeded. The goal of the RL agent is to learn a policy $\pi(a|s)$ that maximizes the expected cumulative reward, corresponding to minimizing infections and improving long-term epidemic outcomes.

V. STATE AND ACTION REPRESENTATION

At each turn, the agent receives information related to the game state, representing the data available to public health authorities. The agent uses this information to inform the state of the game and support which decisions to execute each turn.

A. State Representation

The state vector provided to the agent condenses complex simulator information into a structured feature set. Each state includes:

- **Global epidemic indicators:** active infections, cumulative infections, daily new infections.
- **Community statistics:** infections in each community, capturing geographic clustering and localized outbreaks.
- **Economic indicators:** treasury balance, pending payouts, spending constraints.
- **Vaccine status:** days until approval, dose availability, research progress, distribution capacity.
- **Behavioral metrics:** aggregate risk tolerance, mitigation influence, and adherence levels.

This high-dimensional representation ensures the agent has access to relevant cues that influence disease transmission, resource allocation, and intervention timing.

B. Action Representation

The simulator exposes a discrete action set representing all public-health and economic interventions available to the agent. These include:

- launching awareness or engagement campaigns,
- imposing or relaxing community mobility restrictions,
- allocating vaccine doses through random or referral-based strategies,
- accelerating vaccine production or research,
- adjusting treasury investments or spending.

Because not all actions are legal in every state (e.g., administering vaccines before approval or selecting actions that exceed available funds), action feasibility is explicitly handled

within the Q-learning agent. At each decision step, the agent computes a validity mask based on current financial, epidemiological, and policy constraints, identifying which actions are permissible in the current state. Hence, the learning algorithm operates over a fixed discrete action space while respecting dynamic environment constraints.

VI. REWARD STRUCTURE AND LEARNING ALGORITHM

A. Reward Design

The reward signal is centered on controlling infection spread. At each turn, the agent receives a reward computed as a scaled negative value of new infections. This formulation penalizes actions that accelerate transmission and encourages interventions that delay or suppress outbreaks. Because epidemic control involves inherent trade-offs, this reward structure could be extended to incorporate economic costs or vaccine delays for multi-objective optimization. For the Q-learning implementation used in this work, rewards are not normalized in the same manner as input states. Instead, rewards are divided by a constant factor to account for the fact that multiple actions are taken per decision period before observing the next state.

B. Learning Algorithm

We employ a deep Q-learning approach in which a neural network approximates the action-value function $Q(s, a)$. For the final models evaluated in this study, the Q-network consists of two fully connected hidden layers with 256 and 128 neurons, respectively, followed by a linear output layer corresponding to the discrete action space. LeakyReLU activation functions are used in the hidden layers to improve gradient flow and training stability.

Policy improvement is achieved through an ϵ -greedy exploration strategy. Training begins with high exploration and ϵ decays multiplicatively over time, encouraging exploration early in training and exploitation of learned strategies later. Compared to earlier model variants, the final model uses a faster exploration decay to accelerate convergence toward structured policies.

C. Optimization and Stability Considerations

Training is performed using the Adam optimizer with a fixed learning rate. While learning rate decay was explored, it was not used in the final models due to its decreased performance as the learning rate decreased. Instead, stability is primarily achieved through experience replay and early stopping.

Experiences from many episodes are stored in a replay buffer and sampled uniformly during training, breaking temporal correlations between updates. The replay buffer is periodically truncated to retain recent experiences while maintaining sufficient diversity. Terminal states are handled carefully to prevent bootstrapping beyond episode boundaries. Despite these measures, learning remains highly non-monotonic, motivating the use of evaluation-based early stopping for the final model.

VII. TRAINING METHODOLOGY

All Q-learning models were trained using a common framework in which an agent interacted with the COVID simulation through repeated episodes. Each training epoch consisted of a single full simulation run, after which observed state transitions were appended to a replay buffer and used for batch-wise Q-learning updates. Input states were normalized using a MinMaxScaler fitted on an initial set of random simulations to establish consistent feature scaling. Exploration was governed by an ϵ -greedy policy with multiplicative decay. Periodic evaluation runs were conducted using fully greedy policies and were excluded from the training data.

A. Model 1: Preliminary Draft Model

Model 1 represents an early draft used to validate the simulator-agent interface and the overall training pipeline. This model was not systematically evaluated, as it failed to reduce infections below the random-action baseline. However, it is still included to show our progression throughout the development of the different models.

B. Model 2: Baseline Q-Learning Model

Model 2 employed a deep Q-network with three fully connected hidden layers of sizes 512, 512, and 128, using ReLU activation functions. The agent was trained with a discount factor of $\gamma = 0.9$, a learning rate of 0.00002, a batch size of 128, and an ϵ -decay factor of 0.999 over 2000 training epochs. The replay buffer was periodically truncated to retain the most recent 10,000 transitions. This model demonstrated modest improvements over random action selection and exhibited early signs of structured decision making.

C. Model 3: Extended Horizon and Slower Exploration Decay

Model 3 increased the discount factor to $\gamma = 0.95$ and slowed the exploration decay to 0.9995 while keeping the network architecture unchanged. Training was extended to 4100 epochs, with fully greedy evaluation runs performed periodically to assess performance. Although strong policies emerged early in training, continued optimization led to high variance and eventual degradation of performance.

D. Model 4: Simplified Architecture and Accelerated Training

Model 4 simplified the network architecture to two fully connected hidden layers of sizes 256 and 128, using LeakyReLU activation functions. The discount factor was increased to $\gamma = 0.97$, and the exploration decay was accelerated to 0.997. The batch size was increased to 256, and the replay buffer capacity was expanded to 20,000 transitions. This model achieved the lowest infection counts observed during training but again exhibited instability under prolonged updates.

E. Model 5: Final Early-Stopped Model

Model 5 corresponds to the final model shown in Figure 1. It retained the architecture and discount factor of Model 4 but slightly increased the learning rate to 0.00003 and employed early stopping to preserve strong policies. To increase replay

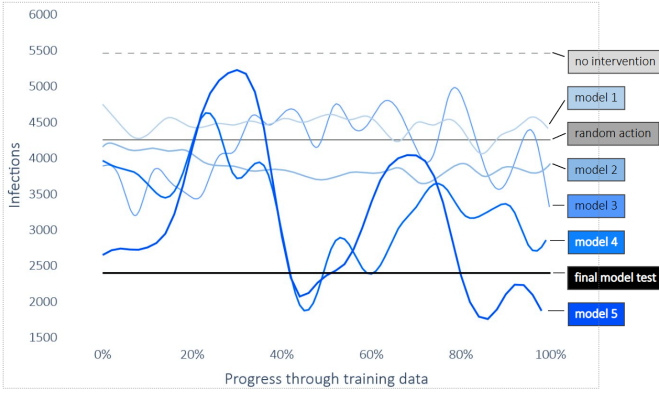


Fig. 1. Average total infections as a function of training progress across Q-learning model variants.

diversity, additional training episodes were generated using randomized exploration levels and inserted into the replay buffer. Training was halted after approximately 200 epochs once consistently strong performance was achieved.

VIII. RESULTS AND VISUALIZATIONS

This section presents the empirical performance of the trained Q-learning agents and analyzes both quantitative outcomes and qualitative policy behavior. Results are organized around three primary visualizations: (i) the number of infections as a function of training progress across model variants, (ii) the evolution of high-level action-selection patterns during gameplay with Model 4, and (iii) the temporal frequency of specific financial and restrictive actions selected by the final trained policy.

A. Training Progress and Model Performance

Figure 1 shows average infections as a function of training progress for multiple Q-learning models, alongside random-action and no-intervention baselines. Learning is highly non-monotonic: early models perform similarly to random, while Model 2 and Model 3 exhibit high variance with alternating improvement and regression. Model 4 achieves a marked reduction in infections around 40% of training, reaching around 2000 average infections before degrading with continued updates. Model 5, the final early-stopped model, preserves this low infection trend by terminating training near the observed minimum.

Table I summarizes performance of the final model over 250 evaluation runs. Across all simulations, average infections are reduced to approximately 2161 compared to 3759 under random action selection, with lower variance. In outbreak-only simulations, infections drop from roughly 4231 (random) to 2388 under the learned policy. Although the learned policy slightly increases average pandemic duration, this reflects delayed and mitigated spread rather than uncontrolled outbreaks. Overall, these results demonstrate substantial performance gains over random baselines and highlight the importance of early stopping in this setting.

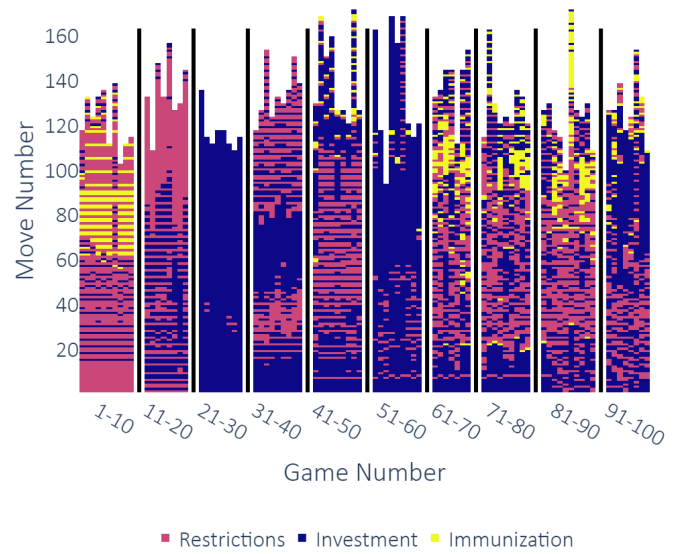


Fig. 2. Distribution of action categories selected by Model 4 over 100 games

B. Learned Policy Structure and Action Selection

Figure 2 visualizes the distribution of action categories selected by the final model over 100 evaluation games. Actions are grouped into three high-level categories: restrictions, investments, and immunization-related interventions. The visualization reveals distinct temporal phases in policy behavior. During early gameplay (approximately games 1–20), the agent predominantly selects simple restriction and income-related actions, reflecting limited information and low infection prevalence. As gameplay progresses (games 21–40), the policy shifts toward investment-heavy strategies, particularly treasury investments and research grants. This transition precedes the strongest performance window.

Between games 41 and 50, which corresponds to the lowest infection counts observed in Figure 1, the agent consistently combines restriction-based actions with awareness campaigns. In later stages (games 61–80), immunization actions emerge more frequently as vaccines become available. While they do appear in mid-to-late gameplay, they do not dominate the final learned policy. Beyond this point (games 81–100), action selection becomes less coherent, with no single dominant pattern, coinciding with declining performance. The emergence and subsequent degradation of structured action sequences provides qualitative evidence that the agent learns non-trivial, context-dependent strategies rather than static or random policies.

Figures 3 and 4 show the frequency of financial and restrictive actions selected by the final model over time, revealing a clear temporal structure in policy behavior. Financial actions exhibit a progression from early treasury investments to community funding and later income-related actions, suggesting that the agent starts off with riskier investments and shifts toward more stable financial decisions as the epidemic evolves. Restrictive actions show a similar pattern: individual pay-

TABLE I
FINAL MODEL PERFORMANCE COMPARED TO RANDOM POLICY (250 SIMULATIONS)

Policy	Scenario	Avg. Infections	Std. Dev.	Pandemic Length
Final RL Model	All simulations	2160.64	825.91	41.38
Final RL Model	Outbreak only (>100)	2388.15	464.05	44.47
Random Policy	All simulations	3758.58	1377.36	36.66
Random Policy	Outbreak only (>100)	4231.24	376.28	39.99

Frequency of Financial Move Selections

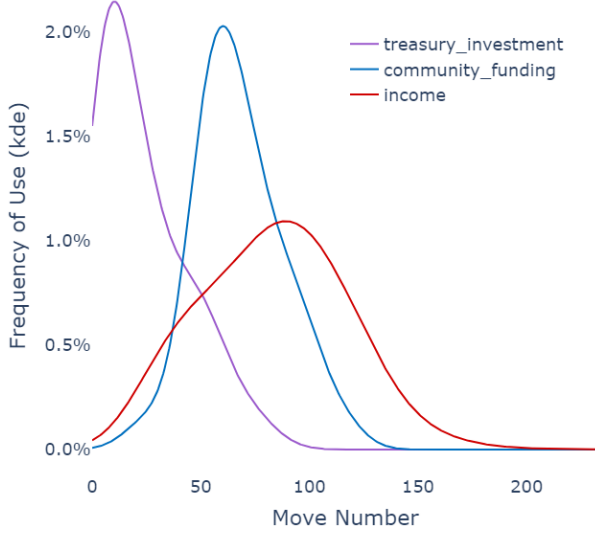


Fig. 3. Frequency of financial action selections

Frequency of Restrictive Move Selections

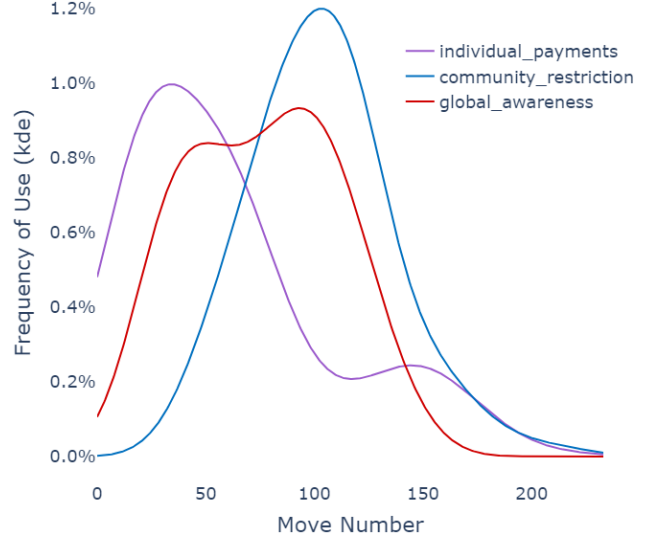


Fig. 4. Frequency of restrictive action selections

ments dominate early gameplay, followed by global awareness campaigns as infections rise, and finally community-specific restrictions in later stages, reflecting increasingly targeted mitigation strategies.

C. Summary of Results

Together, these figures demonstrate that the Q-learning agents can learn effective and structured epidemic intervention strategies within the simulator. Performance improvements emerge early in training and are associated with coherent, multi-stage decision patterns that evolve over time. The action-frequency analyses further show that the final model adopts temporally organized financial and restrictive strategies rather than static or random behavior. However, continued training often leads to policy degradation, motivating the use of early stopping. The final model achieves the strongest and most stable performance, substantially reducing infections relative to random action selection while exhibiting interpretable and intuitive policy structure.

IX. DISCUSSION

The results demonstrate that deep Q-learning can produce meaningful and interpretable epidemic intervention strategies

within a complex, stochastic, agent-based simulation. However, the learning dynamics are highly non-linear and exhibit several patterns of instability and policy degradation with extended training, which are well documented challenges in reinforcement learning [6], [8]. In this section, we analyze the learned behavior of the agent by addressing three key questions.

A. Does the RL Agent Learn Intuitive Strategies?

The learned policies exhibit several behaviors that align closely with intuitive public health decision-making. Early in gameplay, the agent tends to prioritize basic restrictions, income-related actions, and small-scale interventions, reflecting cautious responses when infection prevalence is low and information is limited. As the epidemic progresses, the agent increasingly allocates resources toward investments and awareness campaigns, followed by vaccination strategies once vaccines become available.

These phases closely mirror real world pandemic responses, where governments often begin with containment measures, transition to economic and research investments, and then later rely on vaccination programs once available. Importantly,

these patterns emerge without explicit domain-specific rules encoded into the reward function. This suggests that the agent is learning meaningful relationships between interventions, disease progression, and delayed outcomes rather than following predetermined heuristics.

B. Does the RL Agent Learn Complex Strategies Not Easily Discoverable by Humans?

Beyond intuitive behaviors, the agent also discovers strategies that are more complex and difficult to design manually. High-performing models exhibit coordinated, multi-step decision sequences that exploit delayed rewards. For example, the agent frequently invests early in research grants or treasury growth despite short-term costs, then leverages the resulting financial gains to fund large-scale individual payments or awareness campaigns at later stages of the epidemic.

These strategies require anticipating delayed payoffs under uncertainty and balancing multiple competing objectives, including infection control, economic sustainability, and vaccine development. In a high-dimensional and stochastic environment like our COVID simulation, such planning is not trivial and would be challenging to replicate using hand crafted rules or static heuristics. The emergence of these behaviors highlights the ability of RL to uncover policies that extend beyond straightforward human intuition.

C. Does the RL Agent Improve Consistently Across Training Epochs?

While performance improvements are clearly observed, learning is not consistent or monotonic across training epochs. In several cases, models reach strong performance early in training, only to degrade as training continues. Models that achieve low average infection counts during early or mid-training phases later shift toward less effective strategies, resulting in increased infections.

This behavior is indicative of catastrophic forgetting, a known issue in RL and continual learning where newly acquired knowledge overwrites previously learned policies [6], [7]. The sensitivity of performance to training duration underscores a key challenge in applying deep Q-learning to complex epidemic simulations. Unlike supervised learning, continued training does not guarantee continued improvement. Instead, early stopping emerges as a critical stabilization technique, as demonstrated by the final model, which preserves strong policies by halting training near a performance minimum.

D. Implications for Real-World Epidemic Control

An important observation is that the learned policy slightly increases average pandemic duration while substantially reducing total infections. In the context of real-world epidemic response, this outcome is consistent with the concept of flattening the curve, where transmission is suppressed and spread over a longer time frame to reduce peak burden on healthcare systems. Rather than rapidly exhausting the susceptible population through uncontrolled outbreaks, the agent learns to delay and mitigate transmission through sustained

interventions. This trade-off reflects a desirable public health objective, prioritizing manageable infection levels over rapid epidemic resolution.

Taken all together, these findings highlight both the promise and limitations of RL for epidemic response planning. The agent demonstrates the ability to learn intuitive and complex strategies that significantly reduce infections relative to random baselines, even in the presence of noisy feedback and delayed rewards. At the same time, the instability observed during training emphasizes the importance of careful evaluation, training control, and policy selection.

In real-world public health settings, these results suggest that RL is best suited as a decision-support tool rather than a fully autonomous policymaker. Human oversight, interpretability, and conservative deployment strategies would be essential to ensure reliability and safety when applying RL-based approaches to real epidemic decision making.

X. CONCLUSION AND FUTURE WORK

This work investigated the use of deep Q-learning for epidemic intervention planning within a large-scale, agent-based COVID simulation. By integrating RL with a high-fidelity environment that models individual behavior, social networks, economic decisions, and vaccine development, the study demonstrates that learned policies can substantially reduce infections relative to random baselines while exhibiting interpretable, multi-stage behavior.

Across model variants, effective strategies often emerged early in training but were not preserved under continued optimization. Performance was highly non-monotonic, with policy degradation caused by catastrophic forgetting in later training stages. These findings emphasize the importance of architectural simplicity, exploration control, replay diversity, and early stopping when applying Q-learning in complex, stochastic environments. The final early-stopped model achieved the most stable and robust performance.

A key limitation of the current approach is the many-to-one mapping between multiple actions taken per day and a single aggregated reward, which complicates credit assignment. A promising direction for future work is to redefine the action space at the level of grouped daily interventions, allowing the agent to select a single composite action per day. This would restore a one-action–one-reward structure at the cost of a significantly larger action space (e.g., N^3 combinations for three actions per day). Additional future work includes exploring more stable RL methods such as policy-gradient or actor-critic approaches, richer reward formulations, and techniques to mitigate catastrophic forgetting. These extensions would further improve the reliability of RL as a decision-support tool for epidemic response.

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